

Karnatak University Mathematics Teacher's Association (KUMTA)

B. Sc III Semester (RVSD NEP) Question Bank

Subject: MATHEMATICS. Discipline Specific Core Course (DSC) Course

Title: Algebra - II and Real Analysis – I.

Course Code: C3 MAT 1 T1 (Theory)

Unit I : Groups

15 Hrs.

Syllabus: Groups, Abelian group, Standard examples on a group, Properties of a groups, Subgroup, Permutation group, cyclic groups, Cosets, Lagrange's theorem, Normal subgroups, Quotient groups, Homomorphism and Isomorphism of groups.

Questions carrying Two Marks each

1. Define group .
2. Define abelian group and give an example.
3. Define subgroup and give an example.
4. Prove that identity element in a group is unique.
5. Prove that inverse of every element in a group is unique.
6. Prove that in a group $(G, *)$, $(a^{-1})^{-1} = a$, for every $a \in G$.
7. Prove that in a group $(G, *)$, $(a.b)^2 = a^2.b^2$, for every $a, b \in G$
8. Prove that in a group $(G, *)$,
 $a * b = a * c \Rightarrow b = c$ for every $a, b, c \in G$
9. Prove that a group of order two is abelian.
10. Prove that in a group $(G, *)$ if every element having their own inverse then G is abelian.

OR

Prove that in a group $(G, *)$, if for every $a \in G \Rightarrow a^2 = a$ then G is abelian.

OR

Prove that in a group $(G, *)$, if for every $a \in G \Rightarrow a^{-1} = a$ then G is abelian.

11. Define cyclic group and give an example.
12. Prove that every cyclic group is abelian.
13. If 'a' is a generator of cyclic group G then prove that a^{-1} is also generator of G.
14. Define left and right cosets of H of a group G.
15. Define normal subgroup of a group.
16. Define quotient group.
17. Define homomorphism of a groups,
18. Define isomorphism of a groups,

Questions carrying Five Marks each

II . Groups, Abelian group, Standard examples on a group

19. Prove that $G = \{x + iy ; \forall x, y \in R \text{ and } i = \sqrt{-1}\}$ the set of complex numbers form an abelian group under the addition of complex numbers
20. Show that the set $G = \{a^n / \forall n \in Z\}$ form a group under multiplication.
21. If Q_0 be the set of all rational numbers other than zero and * is a binary operation on Q_0 defined by $a * b = \frac{ab}{5} \quad \forall a, b \in Q_0$. Then prove that $(Q_0, *)$ form an abelian group.
22. Show that the set Q_{-1} of all rational numbers except -1 , with operation defined by $a * b = a + b + ab$, for every $a, b \in Q_{-1}$ is abelian group.
23. Show that the set Q_1 of all rational numbers except 1 , with operation defined by $a * b = a + b - ab$, for every $a, b \in Q_1$ is abelian group.
24. Prove that the set $G = \{1, w, w^2\}$ of all the cube roots of unity forms an abelian group with respect to multiplication.
25. Prove that the set $G = \{1, -1, i, -i\}$ of all the fourth roots of unity forms an abelian group with respect to multiplication.
26. Show that $G = \{0, 1, 2, 3, 4\}$ is a finite abelian group under the addition modulo 5
27. Show that $G = \{2, 4, 6, 8\}$ is a finite abelian group under the multiplication

modulo 10.

II. Properties of a groups

28. Prove that in a group $(G, *)$, $(a * b)^{-1} = b^{-1} * a^{-1}$, for every $a, b \in G$.

29. Prove that in a group $(G, *)$, a solution $a * x = b$ and $y * a = b$ exist and unique.

III. Subgroup

30. Prove that in a group a non empty subset H of a group G is a subgroup of G if and only if (i) $\forall a \in H \Rightarrow a^{-1} \in H$ (ii) $\forall a, b \in H \Rightarrow ab \in H$.

31. Prove that in a group a non empty subset H of a group G is a subgroup of G if and only if $\forall a, b \in H \Rightarrow ab^{-1} \in H$.

32. Prove that intersection of two subgroups is again a subgroup and an example to show that union of two subgroups need not be a subgroup.

33. Prove that the set of square roots of unity is a subgroup of group of fourth roots of unity under the multiplication.

34. Show that set even integers be the subgroup of the group of integers under addition.

IV. Permutation group

35. Prove that the permutations formed by the set $S = \{a, b, c\}$ is a group under the composition of permutations.

V. Cyclic groups

36. Prove that every subgroup H of a cyclic group is also a cyclic group.

37. If G is a cyclic group of order n generated by a . Then prove that a^m generates G if and only if m and n are relatively prime.

38. If $G = \{ \dots, a^{-2}, a^{-1}, e, a^1, a^2, \dots \}$ be an infinite cyclic group generated by a . then prove that

(i) Distinct powers of a are disjoint.

(ii) G has exactly two generators a and a^{-1}

VI. Cosets:

39. If H is a subgroup of G and if $a, b \in G$ then $H_a = H_b$ if and only if $ab^{-1} \in H$
40. If H is a subgroup of G then prove that Two right cosets of H in G either identical or disjoint.
41. If H is a subgroup of G Then prove that all right cosets of H in G have the same number of elements.
42. State and prove Lagrange's theorem of a group

VII. Normal Subgroup:

43. Prove that a subgroup H of a group G is normal if and only if $a^{-1}ha \in H$
 $\forall h \in H$ and $a \in G$.
44. A subgroup H of a group G then prove that $g^{-1}Hg$ is also subgroup of G .
and $a \in G$

VIII. Quotient groups:

45. If G is a group and H is a normal subgroup of G and $xH \cdot yH = xyH$ for all $x, y \in G$ then prove that the set of all left cosets of H in G forms a group under the same operation.
46. Let G be a cyclic group of order 9 generated by a and H be a subgroup of G generated by a^3 then prepare a Cayley table for G/H

IX. Homomorphism and Isomorphism of groups.

47. If $f : G \rightarrow G'$ be a homomorphism where G and G' be two groups.
Then prove that
1. If e is the identity element in G then e' is the identity element of G'
 2. If a^{-1} is the inverse element in G Then $f(a^{-1}) = [f(a)]^{-1} \quad \forall a \in G$.
48. If $f : G \rightarrow G'$ be a homomorphism where G and G' be two groups
then prove that
1. If H is a subgroup of G then $f(H)$ is a subgroup of G'
 2. If H is a cyclic subgroup of G then $f(H)$ is a cyclic subgroup of G'

49. Prove that every infinite cyclic group $G(a)$ is isomorphic to the additive
50. Prove that every cyclic group of order n is isomorphic to the additive group Z_n of integers modulo n .
51. Let $(P, \cdot)$ be a multiplicative group of positive reals and $(R, +)$ be the additive group of real numbers defined by $f(x) = \log_{10} x / \forall x \in P$ then prove that f is homomorphism and check if it is an isomorphism.
52. Let R be the additive group of real numbers $(R, +)$ be the multiplicative group of positive reals then prove that $f : R \rightarrow R^+$ defined $f(x) = e^x \forall x \in R$ is an isomorphism.
53. Let $(R, +)$ be the multiplicative group of positive reals and R be the additive group of real numbers then prove that $g : R \rightarrow R^+$ defined $g(x) = \log x \forall x \in R^+$ is an isomorphism.
54. Let Z be the additive group of integers and E be the additive group of even integers define $f : Z \rightarrow E$ defined $f(x) = 2x \forall x \in Z$ then prove that Z is an isomorphism.

Unit II : Introduction to Real Analysis:

15 Hrs.

Syllabus: Real line, Field and order axioms, Existence of $\sqrt{2}$ Intervals, Bounded and unbounded sets, Supremum and infimum of a set, Completeness property of $\mathbb{R}$, Archimedean property of $\mathbb{R}$. Absolute value of $\mathbb{R}$ – Standard theorems, Neighbourhood of a point, Interior point of a set, Open set, closed set, Limit points of a set, Bolzano-Weierstrass theorem (statement only), Countable and uncountable sets – standard theorems.

Questions carrying Two Marks each

1. Define bounded and unbounded set.
2. Define infimum and supremum of a set.
3. State Trichotomy law of order axioms.
4. State Transitive law of order axioms.
5. Using field axioms of real numbers show that, $0x = 0$.
6. Using field axioms of real numbers Show that $ab = 0 \Rightarrow$ either $a = 0$ or $b = 0$.

7. Using field axioms of real numbers Show that $x + y = x + z \Rightarrow y = z$
8. Using field axioms of real numbers show that $xy = xz \Rightarrow y = z$.
9. Prove that $|x \cdot y| = |x| \cdot |y|$
10. Define neighbourhood of a point.
11. Define interior point of a set.
12. Define limit point of a set.
13. State Bolzano – Weirstrass theorem.
13. Define Countable set and give an example.

Questions carrying Five Marks each

14. State and prove Archimedean property of a real number $\mathbb{R}$.
15. If $\forall x, y, z \in \mathbb{R}$ then show that $x^2 + y^2 + z^2 \geq xy + yz + zx$.
16. If a, b, c are all positive real numbers $\mathbb{R}$ then show that $\frac{a + b + c}{3} \geq \sqrt[3]{abc}$.
17. Prove that $|x + y| \leq |x| + |y|$
18. Prove that $|x - y| \geq ||x| - |y||$
19. Show that $|a + b|^2 + |a - b|^2 = 2|a|^2 + 2|b|^2$
20. Prove that any open interval (a, b) is a neighbourhood of each of its points
21. Prove that the closed set $[a, b]$ is a neighbourhood of each of its points except a and b .
22. Prove that the intersection of finite number of neighborhood of a point x is also a neighbourhood of the point.
23. If A is countable infinitely and B is infinite subset of A then prove that B is countable.
24. Prove that the union of two countable set is countable
25. Prove that the set $\mathbb{N} \times \mathbb{N}$ is countable.
26. Prove that the set of rational numbers $\mathbb{Q}$ is countable (denumerable).
27. Prove that $[0, 1]$ is uncountable.

Unit III: Sequences

15 Hrs.

Sequences of real numbers, Bounded sequences, Limit of a sequence, Convergent, Divergent, and Oscillatory sequences, Monotonic sequences, Algebra of convergent sequences, Limit points of a sequence, Bolzano - Weierstrass theorem for sequence, Cauchy's first and second theorem on limits of a sequence, Cauchy's general principle for convergence of a sequence (Statement).

I. Questions carrying Two Marks each

1. Define limit of a sequence .
2. Define convergent sequence and give an example.
3. Define divergent sequence and give an example.
4. Define oscillatory sequence and give an example.
5. Define monotonic increasing and decreasing sequence.
6. If $\lim_{n \rightarrow \infty} x_n = l$ then prove that $\lim_{n \rightarrow \infty} |x_n| = |l|$
7. Prove that every convergent sequence is bounded
8. Prove that limit of convergent sequence is unique.
9. Show that $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$.
10. Show that the sequence $\left\{ \frac{n}{n+1} \right\}_{n \in \mathbb{N}}$ is monotonically increasing .
11. Show that the sequence $\left\{ \frac{n+1}{n} \right\}_{n \in \mathbb{N}}$ is monotonically decreasing.
12. State the Cauchy's first theorem on limit of a sequence .
13. State the Cauchy's second theorem on limit of a sequence .
14. Show that $\lim_{n \rightarrow \infty} \left[\frac{1}{n} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) \right] = 0$.
15. Show that $\lim_{n \rightarrow \infty} \left[\frac{1}{n} \left(\frac{2}{1} + \frac{3}{2} + \frac{4}{3} - \dots + \frac{n}{n+1} \right) \right] = 1$.
16. Show that $\lim_{n \rightarrow \infty} \left[\frac{1}{n} \left(\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} - \dots + \frac{1}{n^2} \right) \right] = 0$.
17. Show that $\lim_{n \rightarrow \infty} \left[\frac{1}{n} \left(1 + \frac{1}{2^{\frac{1}{2}}} + \frac{1}{3^{\frac{1}{3}}} + \dots + \frac{1}{n^{\frac{1}{n}}} \right) \right] = 1$.

18. Show that $\lim_{n \rightarrow \infty} \left(\frac{n^n}{n!} \right)^{\frac{1}{n}} = e$.
19. State Bolzano Weirstrass theorem of a sequence.
20. State Cauchy's general principle of convergence.
21. Show that every Cauchy's sequence is bounded.
22. Show that the sequence $\left\{ \frac{1}{n} \right\}_{n \in \mathbb{N}}$ is Cauchy's sequence.

Question Carrying five marks each

II . Algebra of Limits

- 23 . if $\lim_{n \rightarrow \infty} x_n = l$ and $\lim_{n \rightarrow \infty} y_n = m$
 then prove that $\lim_{n \rightarrow \infty} (x_n + y_n) = l + m$
- 24 . if $\lim_{n \rightarrow \infty} x_n = l$ and $\lim_{n \rightarrow \infty} y_n = m$
 then prove that $\lim_{n \rightarrow \infty} (x_n - y_n) = l - m$
- 25 . if $\lim_{n \rightarrow \infty} x_n = l$ and $\lim_{n \rightarrow \infty} y_n = m$
 then prove that $\lim_{n \rightarrow \infty} (x_n \cdot y_n) = l \cdot m$
- 26 . if $\lim_{n \rightarrow \infty} x_n = l$ and $\lim_{n \rightarrow \infty} y_n = m$
 then prove that $\lim_{n \rightarrow \infty} \left(\frac{x_n}{y_n} \right) = \frac{l}{m}$

III. Cauchy's Theorems on Limit of a sequence

27. State and prove Cauchy's first theorem on limit of a sequence .

OR

If the Sequence $\{x_n\}$ converges to l , then prove that the sequence $\{a_n\}$ also converges to l . Where $a_n = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$

that is a_n is the nth arithmetic mean of $\{x_n\}$

28. State and prove Cauchy's second theorem on limit of a sequence .

OR

If $\langle x_n \rangle$ is a sequence of positive terms a $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$ exists whether finite or infinite

$$\text{then } \lim_{n \rightarrow \infty} (x_n)^{1/n} = \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$$

29. Using Cauchy's first theorem on limits, show that

$$\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2 + 1}} + \frac{1}{\sqrt{n^2 + 2}} + \frac{1}{\sqrt{n^2 + 3}} - - - - - + \frac{1}{\sqrt{n^2 + n}} \right] = 1.$$

30. If $a_n > 0, \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = l$

$$\text{then prove that } \lim_{n \rightarrow \infty} (a_1 \cdot a_2 \cdot a_3, - - - - - a_n)^{\frac{1}{n}} = l$$

$$31. \text{ If } a_n = \left[\left(\frac{2}{1}\right)^1 \cdot \left(\frac{3}{2}\right)^2 \cdot \left(\frac{4}{3}\right)^3 \dots \dots \dots \left(\frac{n+1}{n}\right)^n \right]^{\frac{1}{n}}$$

$$\text{then prove that } \lim_{n \rightarrow \infty} a_n = e.$$

$$32. \text{ Prove that } \lim_{n \rightarrow \infty} \left(\frac{n^n}{n!}\right)^{\frac{1}{n}} = e$$

$$33. \text{ prove that the sequence } \left[\frac{(3n)!}{(n!)^3} \right]^{\frac{1}{n}} \text{ is convergent.}$$

$$34. \text{ prove that the sequence } \lim_{n \rightarrow \infty} \left[\frac{(n+1)(n+2) \dots \dots \dots (n+n)}{n^n} \right]^{\frac{1}{n}} \text{ is convergent.}$$

IV. Monotonic sequence.

34. Prove that a monotonic increasing sequence which is bounded above is always convergent and Type equation here.it converges to its supremum.

35. Prove that a monotonic decreasing sequence which is bounded below is always convergent and it converges to its infimum.

36. Discuss the convergence of the sequence whose n^{th} term is given by

$$x_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n}.$$

37, Discuss the convergence of the sequence whose n^{th} term is given by

$$x_n = \frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1}.$$

38. Discuss the convergence of the sequence whose n^{th} term is given by

$$x_n = \frac{1}{1+1^2} + \frac{1}{1+2^2} + \frac{1}{1+3^2} + \dots + \frac{1}{1+n^2}.$$

39. Discuss the convergence of the sequence whose n^{th} term is given by

$$x_n = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n.n+1}.$$

40. Discuss the convergence of the sequence whose n^{th} term is given by

$$x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}.$$

41. Prove that the sequence $\{x_n\}$ defined by $x_1 = 1$ and $x_n = \sqrt{2 + x_{n-1}}$

$\forall n \geq 2$ is convergent and it converges to 2.

42. Prove that the sequence $\{x_n\}$ defined by $x_1 = \sqrt{6}$ and $x_n = \sqrt{6 + x_n}$

$\forall n \geq 1$ is convergent and it converges to 3.

43. Prove that the sequence $\{x_n\}$ defined by $x_1 = \sqrt{5}$ and $x_n = \sqrt{5x_n}$

$\forall n \geq 1$ is convergent and it converges to 5.

43. Prove that the sequence $\{x_n\}$ defined by $x_1 = \sqrt{2}$ and $x_n = \sqrt{2x_n}$

$\forall n \geq 1$ is convergent and it converges to 2.

Unit IV: Infinite Series

15 Hrs.

Definition of convergent, divergent, and oscillatory series, Geometric series, Cauchy's general principle of convergence of series, p-series, Comparison tests for positive term series. D'Alembert's ratio test, Raabe's test, Cauchy's Root test and Cauchy's integral test, Alternating series, Leibnitz's theorem, Absolute convergence and conditional convergence of a series.

I. Questions carrying Two Marks each

1. State geometric series test

2. Show that the series $1 + \frac{1}{2} + \frac{1}{3} + \dots$ is divergent.

3. Show that the series $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$ is convergent.

6. Show that the series $1 + \frac{1}{\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2}} \pm \dots$ is convergent.

4. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

2. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2 \log n}$

3. Test the convergence of the series $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$

3. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{1}{n}\right)$

4. Test the convergence of the series, $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

5. Show that the series $1 + \frac{1}{4^{\frac{1}{3}}} + \frac{1}{9^{\frac{1}{3}}} + \frac{1}{16^{\frac{1}{3}}} \pm \dots$ is convergent.

7. Show that the series $\frac{\sqrt{2}}{3\sqrt{4}} + \frac{\sqrt{3}}{4\sqrt{7}} + \frac{\sqrt{4}}{5\sqrt{10}} + \dots$ is divergent.

8. Show that $\sum (\sqrt{n+1} - \sqrt{n})$ is divergent

9. If the series $\sum_{n=1}^{\infty} u_n$ is convergent then Show that $\lim_{n \rightarrow \infty} u_n = 0$
10. State D'Alembert's ratio test for series of positive terms.
11. State Cauchy's root test for series of positive terms.
12. State Raabe's ratio test for series of positive terms.
14. State Cauchy's integral test for series of positive terms.
15. State Leibnitz's test for an alternating series.
16. Test for convergence of the series, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$
17. Test for convergence of the series, $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$
18. Define a conditionally convergent series and give an example.
19. Define a absolute convergent series and give an example,

Question Carrying five marks each

II. Geometric series

20. Prove that the geometric series

$$\sum_{n=1}^{\infty} r^{n-1} = 1 + r + r^2 + r^3 + \dots + r^{n-1} + \dots \text{ is}$$

- i) convergent if $-1 < r < 1$ or $|r| < 1$ (ii) divergent if $r \geq 1$ and
 iii) oscillates if $r \leq -1$.

Cauchy's general principle of convergence of series

21. State and prove Cauchy's general principle of convergence of series

II. Comparison Tests of infinite series

22. If $\sum a_n$ and $\sum b_n$ be the series of Positive terms and $a_n \leq b_n$
 and If $\sum b_n$ converges then prove that $\sum a_n$ also converge .
23. If $\sum a_n$ and $\sum b_n$ be the series of Positive terms and $a_n \geq b_n$
 and If $\sum b_n$ diverges then prove that $\sum a_n$ also diverge .

24. If $a_n > 0$ and $b_n > 0$ and $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = l \ (\neq 0)$ then prove that

$\sum a_n$ and $\sum b_n$ behave alike .

P – Series Test

25. Prove that the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$

is (i) convergent if $p > 1$ and (ii) divergent if $p \leq 1$.

OR

State and prove P – series test for the infinite series

III . Discuss the convergence and divergence of the following series

26. $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$

27. $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$

28. $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \dots$

29. $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots$

30. $\frac{1}{\sqrt{1.2}} + \frac{1}{\sqrt{2.3}} + \frac{1}{\sqrt{3.4}} + \dots$

31. $\frac{\sqrt{2} - 1}{3^3 - 1} + \frac{\sqrt{3} - 1}{4^3 - 1} + \frac{\sqrt{2} - 1}{5^3 - 1} + \dots$

IV . Discuss the convergence and divergence of the following series

32. $\sum (\sqrt{n^2 + 1} - \sqrt{n^2 - 1})$

33. $\sum (\sqrt{n^3 + 1} - \sqrt{n^3 - 1})$

34. $\sum (\sqrt{n+1} - \sqrt{n+1})$

35. $\sum (\sqrt{n^4 + 1} - \sqrt{n^4 - 1})$

36. $\frac{\sqrt{2} - \sqrt{1}}{1} + \frac{\sqrt{3} - \sqrt{2}}{2} + \frac{\sqrt{4} - \sqrt{3}}{3} + \dots$

37. State and prove D'Almbert's Ratio test for the infinite series .

38. State and prove Raabe's test for the infinite series .

V. Discuss the convergence of the follwings

$$39. \quad \frac{1}{3} + \frac{1 \cdot 2}{3 \cdot 5} + \frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{3 \cdot 5 \cdot 7 \cdot 9} + \dots$$

$$40. \quad \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 5 \cdot 8} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 5 \cdot 8 \cdot 11} + \dots$$

$$41. \quad \frac{5}{2+5} + \frac{5^2}{2^2+5} + \frac{5^2}{2^3+5} + \dots$$

$$42. \quad \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots$$

VI. Discuus the convergence and divergence of the following series

$$43. \quad \sum \frac{3^n \cdot n!}{n^n}$$

$$44. \quad \sum \frac{2^n \cdot n!}{n^n}$$

$$45. \quad \sum \frac{x^n}{n^2}$$

$$46. \quad \sum \frac{x^n}{n(n+1)}$$

$$47. \quad \frac{x}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^2}{4\sqrt{5}} + \dots$$

VI. Discuus the convergence and divergence of the following series

$$48. \quad \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \dots$$

$$49. \quad \frac{1^2}{2^2} + \frac{1^2 \cdot 2^2}{2^2 \cdot 4^2} + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

$$50. \quad \frac{1^2}{4^2} + \frac{1^2 \cdot 5^2}{4^2 \cdot 8^2} + \frac{1^2 \cdot 5^2 \cdot 9^2}{4^2 \cdot 8^2 \cdot 12^2} + \dots$$

$$51. \quad \frac{1^2}{2^2} + \frac{1^2 \cdot 2^2}{2^2 \cdot 4^2} x + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} x^2 + \dots$$

52. State and prove Cauchy's root test fot the infinite series

VIII. Discuus the convergence and divergence of the following series

$$53. \quad \left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3} + \dots$$

$$54. \quad \frac{2}{1^2} x + \frac{3^2}{2^3} x^2 + \frac{4^3}{3^4} x^3 + \dots$$

OR

$$\sum \frac{(n+1)^n \cdot x^n}{n^{n+1}}$$

55. $\sum \left(1 + \frac{1}{n}\right)^{-n^2}$

56. $\sum \left(\frac{n}{n+1}\right)^{n^2}$

57. State and prove Cauchy's Integral test for the infinite series

IX. Discuss the convergence and divergence of the following series

58. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

59. $\sum_{n=1}^{\infty} \frac{2n^3}{n^4 + 3}$

60. $\sum_{n=1}^{\infty} \frac{n}{(n^2 + 1)^2}$

61. $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2 - 1}}$

62. $\sum_{n=1}^{\infty} \frac{1}{(n+1)\log(n+1)}$

63. $\sum_{n=1}^{\infty} \frac{1}{n(\log n)^2}$

64. State and prove Leibnitz's test on alternating series

65. $\frac{1}{1.3} - \frac{1}{2.4} + \frac{1}{3.5} - \frac{1}{4.6} - - - - -$

66. $\frac{1}{\log 2} - \frac{1}{\log 3} + \frac{1}{\log 4} - \frac{1}{\log 5} - - - - -$

XI. Test the convergence and absolute convergence of the series

67. $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + - - - - -$

68. $2 - \frac{3}{2} + \frac{4}{3} - \frac{5}{4} + - - - - -$

69. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)!}$

70. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n\sqrt{n}}$

71. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \cdot n}{(2n-1)!}$

Unit IV: Infinite Series

15 Hrs.

Definition of convergent, divergent, and oscillatory series, Geometric series, Cauchy's general principle of convergence of series, p-series, Comparison tests for positive term series. D'Alembert's ratio test, Raabe's test, Cauchy's

Root test and Cauchy's integral test, Alternating series, Leibnitz's theorem, Absolute convergence and conditional convergence of a series.

I. Questions carrying Two Marks each

1. State geometric series test
2. Show that the series $1 + \frac{1}{2} + \frac{1}{3} + \dots$ is divergent.
3. Show that the series $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$ is convergent.
6. Show that the series $1 + \frac{1}{\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2}} \pm \dots$ is convergent.
4. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$
2. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2 \log n}$
3. Test the convergence of the series $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$
3. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{1}{n}\right)$
4. Test the convergence of the series, $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$
5. Show that the series $1 + \frac{1}{4^{\frac{1}{3}}} + \frac{1}{9^{\frac{1}{3}}} + \frac{1}{16^{\frac{1}{3}}} \pm \dots$ is convergent.
7. Show that the series $\frac{\sqrt{2}}{3\sqrt{4}} + \frac{\sqrt{3}}{4\sqrt{7}} + \frac{\sqrt{4}}{5\sqrt{10}} + \dots$ is divergent.
8. Show that $\sum (\sqrt{n+1} - \sqrt{n})$ is divergent
9. If the series $\sum_{n=1}^{\infty} u_n$ is convergent then Show that $\lim_{n \rightarrow \infty} u_n = 0$
10. State D'Alembert's ratio test for series of positive terms.
11. State Cauchy's root test for series of positive terms.

12. State Raabe's ratio test for series of positive terms.
14. State Cauchy's integral test for series of positive terms.
15. State Leibnitz's test for an alternating series.
16. Test for convergence of the series, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$
17. Test for convergence of the series, $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$
18. Define a conditionally convergent series and give an example.
19. Define an absolute convergent series and give an example,

Question Carrying five marks each

II. Geometric series

20. Prove that the geometric series

$$\sum_{n=1}^{\infty} r^{n-1} = 1 + r + r^2 + r^3 + \dots + r^{n-1} + \dots \text{ is}$$

- i) convergent if $-1 < r < 1$ or $|r| < 1$ (ii) divergent if $r \geq 1$ and
- iii) oscillates if $r \leq -1$.

Cauchy's general principle of convergence of series

21. State and prove Cauchy's general principle of convergence of series

II. Comparison Tests of infinite series

22. If $\sum a_n$ and $\sum b_n$ be the series of Positive terms and $a_n \leq b_n$

and If $\sum b_n$ converges then prove that $\sum a_n$ also converge.

23. If $\sum a_n$ and $\sum b_n$ be the series of Positive terms and $a_n \geq b_n$

and If $\sum b_n$ diverges then prove that $\sum a_n$ also diverge.

24. If $a_n > 0$ and $b_n > 0$ and $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = l \ (\neq 0)$ then prove that

$\sum a_n$ and $\sum b_n$ behave alike.

P – Series Test

25. Prove that the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$

is (i) convergent if $p > 1$ and (ii) divergent if $p \leq 1$.

OR

State and prove P – series test for the infinite series

III . Discuss the convergence and divergence of the following series

26. $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$

27. $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$

28. $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \dots$

29. $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots$

30. $\frac{1}{\sqrt{1.2}} + \frac{1}{\sqrt{2.3}} + \frac{1}{\sqrt{3.4}} + \dots$

31. $\frac{\sqrt{2} - 1}{3^3 - 1} + \frac{\sqrt{3} - 1}{4^3 - 1} + \frac{\sqrt{2} - 1}{5^3 - 1} + \dots$

IV . Discuss the convergence and divergence of the following series

32. $\sum (\sqrt{n^2 + 1} - \sqrt{n^2 - 1})$

33. $\sum (\sqrt{n^3 + 1} - \sqrt{n^3 - 1})$

34. $\sum (\sqrt{n + 1} - \sqrt{n + 1})$

35. $\sum (\sqrt{n^4 + 1} - \sqrt{n^4 - 1})$

36. $\frac{\sqrt{2} - \sqrt{1}}{1} + \frac{\sqrt{3} - \sqrt{2}}{2} + \frac{\sqrt{4} - \sqrt{3}}{3} + \dots$

37. State and prove D'Almbert's Ratio test for the infinite series .

38. State and prove Raabe's test for the infinite series .

V. Discuss the convergence of the followings

39. $\frac{1}{3} + \frac{1.2}{3.5} + \frac{1.2.3}{3.5.7} + \frac{1.2.3.4}{3.5.7.9} + \dots$

40. $\frac{1}{2} + \frac{1.3}{2.5} + \frac{1.3.5}{2.5.8} + \frac{1.3.5.7}{2.5.8.11} + \dots$

$$41. \frac{5}{2+5} + \frac{5^2}{2^2+5} + \frac{5^2}{2^3+5} + \dots$$

$$42. \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots$$

VI. Discuss the convergence and divergence of the following series

$$43. \sum \frac{3^n \cdot n!}{n^n}$$

$$44. \sum \frac{2^n \cdot n!}{n^n}$$

$$45. \sum \frac{x^n}{n^2}$$

$$46. \sum \frac{x^n}{n(n+1)}$$

$$47. \frac{x}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^2}{4\sqrt{5}} + \dots$$

VI. Discuss the convergence and divergence of the following series

$$48. \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \dots$$

$$49. \frac{1^2}{2^2} + \frac{1^2 \cdot 2^2}{2^2 \cdot 4^2} + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

$$50. \frac{1^2}{4^2} + \frac{1^2 \cdot 5^2}{4^2 \cdot 8^2} + \frac{1^2 \cdot 5^2 \cdot 9^2}{4^2 \cdot 8^2 \cdot 12^2} + \dots$$

$$51. \frac{1^2}{2^2} + \frac{1^2 \cdot 2^2}{2^2 \cdot 4^2} x + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} x^2 + \dots$$

52. State and prove Cauchy's root test for the infinite series

VIII. Discuss the convergence and divergence of the following series

$$53. \left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3} + \dots$$

$$54. \frac{2}{1^2} x + \frac{3^2}{2^3} x^2 + \frac{4^3}{3^4} x^3 + \dots$$

OR

$$\sum \frac{(n+1)^n \cdot x^n}{n^{n+1}}$$

$$55. \sum \left(1 + \frac{1}{n}\right)^{-n^2}$$

$$56. \sum \left(\frac{n}{n+1}\right)^{n^2}$$

57. State and prove Cauchy's Integral test for the infinite series

IX. Discuss the convergence and divergence of the following series

58. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

59. $\sum_{n=1}^{\infty} \frac{2n^3}{n^4 + 3}$

60. $\sum_{n=1}^{\infty} \frac{n}{(n^2 + 1)^2}$

61. $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2 - 1}}$

62. $\sum_{n=1}^{\infty} \frac{1}{(n+1)\log(n+1)}$

63. $\sum_{n=1}^{\infty} \frac{1}{n(\log n)^2}$

64. State and prove Leibnitz's test on alternating series

65. $\frac{1}{1.3} - \frac{1}{2.4} + \frac{1}{3.5} - \frac{1}{4.6} - - - - -$

66. $\frac{1}{\log 2} - \frac{1}{\log 3} + \frac{1}{\log 4} - \frac{1}{\log 5} - - - - -$

XI. Test the convergence and absolute convergence of the series

67. $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + - - - - -$

68. $2 - \frac{3}{2} + \frac{4}{3} - \frac{5}{4} + - - - - -$

69. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)!}$

70. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n\sqrt{n}}$

71. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \cdot n}{(2n-1)!}$

Marks Distribution				
Units	I. Two marks	II. Five marks	III. Ten marks	Total marks
Groups	2	2	2	24
Introduction to Real Analysis:	2	2	2	24
Sequences	3	2	2	26
Infinite Series	3	2	2	26
	10	08	04	110

Question Paper carrying 80 marks

First Main: (Two marks each, $2 \times 10 = 20$) Answer **all** questions.

Second Main: (Five marks each, $5 \times 6 = 30$) Answer any **six** out of **eight**,

Third Main: (Ten marks each, $10 \times 3 = 30$) Answer any **Three** out of **four**